

Maximo Babun

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EDUCATION

University of San Diego

*B.S. Computer Science concentration on Data Science and AI, Minor Management
Trustee Scholarship and USD Grant — 3.87 — Dean's First Honors*

San Diego, CA

Aug. 2023 – May 2027

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML/CSS, SQL

Developer Tools: Git, VS Code, Cursor, IntelliJ, Linux/Bash, VIM/NVIM, MySQL, PostgreSQL

Libraries/ Frameworks: Pandas, Matplotlib, JUnit, Mockito, Docker, Spring

Related Concepts: Object Oriented Programming, Database Management, Prompt Engineering

PROJECTS

Spotify Popularity and Genre Prediction | *Python, Pandas, Scikit-learn, Matplotlib*

Nov. - Dec. 2025

- Designed and implemented a full machine learning workflow on 232k+ Spotify tracks using Python, pandas, and scikit learn, including data cleaning, feature engineering, correlation analysis, and preprocessing of audio features.
- Collaborated on a team to build and evaluate a logistic regression model for hit song prediction, addressing class imbalance with class weighting and stratified cross validation, achieving ROC AUC = 0.95 while analyzing **precision recall tradeoffs**.
- Applied **K Means clustering** to discover interpretable song groups using standardized audio features and PCA visualization, then trained a multi label **XGBoost classifier** for genre prediction, evaluating per genre F1 scores and identifying limitations of audio only genre classification.

Command Interpreter Shell | *C*

April 2024

- Architected a Unix shell in **C** featuring dynamic memory allocation for command parsing with robust input tokenization and background process execution.
- Implemented comprehensive **signal handling** system for process control, enabling foreground process interruption and asynchronous background process monitoring.
- Engineered **process management** architecture with child process creation and collection to prevent zombie processes while tracking exit codes and termination signals.
- Developed built-in command functionality including directory navigation with environment variable integration and comprehensive error handling throughout the execution.

EXPERIENCE

Lab Assistant and Tutor

Shiley-Marcos School of Engineering

February 2024 – Present

San Diego, CA

- Provide technical guidance to 100+ students in programming concepts, including data structures, algorithm analysis, and **debugging** in Python, Java, and C.
- Mentor students on **object-oriented programming**, functional programming, abstraction techniques, and low-level computing concepts such as memory organization and concurrency.
- Lead one-on-one and group tutoring sessions on **software development best practices** and **coding standards**, increasing student engagement and assignment quality while improving programming efficiency.

Software Engineering Intern (Incoming)

General Atomics Aeronautical Systems

May 2026 – Aug. 2026

Poway, CA

ORGANIZATIONS

Association for Computing Machinery (ACM) – Vice President, 2025–2026

Theta Tau – Professional Engineering Fraternity – 2025 Scribe, 2026 Web Page and Service Chair

Society of Hispanic Professional Engineers (SHPE) - Active Member